

Linke Fan

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Education Background

Tongji University (Project 985/211 university)

Sep. 2023 - Jun. 2027

- Degree: Bachelor of Engineering in Computer Science and Technology
- GPA: 85.04 /100
- Language Skills: English (IELTS: 7.5); Chinese (native)
- Relevant Coursework:
 - AI & Data Science: Machine Learning (B), Artificial Intelligence (B), Chinese Information Processing (A)
 - Systems & Algorithms: Compilers (A), Computer Organization (A), Algorithm Design and Analysis (B)
 - Software & Practicums: Database System Practicum (A), Operating System Practicum (A), Human Computer Interaction (A)

Academic Publication

FSB-Net: Frequency-Spatial Boundary Network for Brain Stroke Lesion Segmentation in Non-Contrast CT

Jan. 2026 - Jul. 2026

Linke Fan[†], Xianglong Li[†], Huixin Huang, and Kai Shu* ([†]Equal contribution, *Corresponding author)

Accepted for publication in *Proceedings of the 2026 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Dec. 2026. Paper ID: S13202. arXiv:2607.20955. [CCF-B conference]

- Network Architecture Design:** Architected the overall network architecture and implemented FSB-Net, a frequency-spatial fusion lesion segmentation network; introduced wavelet transform to extract high-frequency boundary information and designed a cross-attention mechanism for bidirectional information interaction between frequency and spatial domains
- Wavelet Boundary Detection Head (WBDH) Design:** Designed the WBDH using DWT to multi-scale encoder feature maps to decompose them into low-frequency approximation and horizontal/vertical/diagonal high-frequency details, explicitly modeling latent boundaries, and adding learnable threshold gating to suppress high-frequency responses with low amplitudes
- Frequency-Spatial Cross-Attention (FSCAM) Design:** Designed the FSCAM by using spatial features as Query and frequency-domain boundary features as Key/Value for weighted fusion, updating frequency-domain features with spatial features as reference, and adopting a training strategy in which spatial features dominated and bidirectional interaction was gradually enabled after training stabilized
- Spectral Boundary Loss (SBL) Design:** Formulated SBL to compute high-frequency component discrepancies between predicted segmentation maps and ground-truth labels in the Fourier domain, and combined Dice/BCE, SSIM, deep supervision, and an FPN decoder for training and auxiliary supervision
- Experiments and Validation:** Evaluated morphological gradient, Canny, HED-style side outputs, and Haar DWT; built a PVTv2 encoder-decoder internal baseline and incrementally added WBDH, FSCAM, SBL, and deep supervision; verified that bidirectional cross-attention outperformed concatenation, addition, residual gating, and unidirectional attention, and that SBL and SSIM were complementary

Internship Experience

JD Technology (JD Cloud) | Software Development Intern

Aug. 2025 - Sep. 2025

Core project: End-to-end Data Migration for the JD Cloud Data Lake Project

- Migration Scripts:** Developed automated migration scripts to handle differences between source and target in table schemas, Parquet/ORC data formats, partitioning strategies, and storage engines; completed three-level organizational mapping from source table schemas to target Catalog/Schema/Table, redesigned partition keys, and performed lossless data format conversion; successfully migrated 80+ tables (30TB+ data) with 100% data consistency, supporting 20+ downstream business tasks
- Spark/Hive Performance Tuning:** Tuned Spark/Hive performance by reading Spark and Hive execution plan logs and analyzing slow-query bottlenecks using EXPLAIN; addressed data skew in aggregation queries that overloaded a single Reducer by adjusting bucketing keys and introducing random prefixes to scatter hot keys, reducing query time from over 3 minutes to under 20 seconds (a ~90% reduction) and cutting cluster resource consumption
- Cloud Environment Setup and Data Validation:** Configured cloud environments by setting up runtime environments, creating databases and tables, and executing queries on cloud servers to verify data integrity; coordinated permission management, security groups, and VPC network policies, including network ACLs, sub-user authorization, user groups, and role authorization, to ensure migration link connectivity and data correctness
- Fault Localization:** Diagnosed migration failures by troubleshooting layer by layer across source-side network, target-side permissions, and script logic; documented and reported problem symptoms and workarounds to the data source team, target platform operations team, and business acceptance team; generated traceable logs for dependencies to improve traceability, verifiability, and collaboration in the migration process

Beijing Qihu Ceteng Security Technology Co., Ltd. | Pre-sales Technical Intern

Jul. 15, 2025 - Jul. 31, 2025

- Technical Solution Design:** Participated in customer interviews and translated business requirements (intranet security, MLPS compliance, data leakage) into technical specifications for DLP endpoint control, UEBA behavior analytics, and database auditing; designed tailored security solutions for 5 client scenarios.

- **Competitive Analysis & Communication:** Compiled 10+ competitive analysis reports and technical comparison matrices to support proposal writing; bridged technical terminology with user-friendly language for customer-facing presentations

CCB Fintech Co., Ltd. | Production and Technology Committee Office Intern **Jul. 1, 2024 - Jul. 31, 2024**

- **Root Cause Analysis:** Investigated and documented 30+ production security incidents; traced the complete causal chain from technical phenomena to business impact by reviewing change records, confirming operational details with developers, and inspecting business monitoring dashboards
- **System Reliability & Solution Review:** Engaged in 10+ core system proposal reviews and contributed to improving the production safety retrospective system; gained hands-on exposure to the financial-grade system reliability closed loop, including design review, canary release, monitoring/alerting, and emergency rollback

Scientific Research

CausIDL: Resource Competition-Aware Causal Graph Discovery for Deep Learning Microservices
Individual Research Project | Supervised by: Professor Zhijun Ding **Sep. 2025 - Jun. 2026**

- **Instance-Level Modeling:** Modeled causality at the individual service instance level by incorporating resource metrics such as GPU/CPU utilization, memory bandwidth, and GPU memory usage, as well as performance metrics such as request volume, error rate, and latency, avoiding the instance homogeneity assumption at service-type granularity
- **Competition-Aware Hierarchical Modeling:** Introduced node-level aggregate metrics as shared parents to encode indirect resource competition among instances sharing a physical node as mediation paths, distinguishing application-layer calls from physical-layer resource competition and avoiding misattribution as direct causality between instances
- **Domain Knowledge Constraints:** Encoded GPU hardware characteristics, container isolation principles, and inference service workflows into a forbidden edge list, reducing the causal search space by 20%+ and improving graph accuracy; excluded physically invalid edges such as performance-to-resource and instance-to-node-level directions
- **Semi-Automated Validation:** Built semi-automated validation to verify algorithm outputs against hardware logic and service workflows, identifying 10+ anomalous causal edges on average; added violating edges to the forbidden list and reran the algorithm to iteratively improve graph quality

Serverless End-to-End Automated AI Workflow System
Individual Research Project | Supervised by: Professor Zhijun Ding **Dec. 2023 - Dec. 2024**

- **System Architecture Design:** Designed the overall architecture of a serverless AI workflow platform based on Knative and Argo, and developed an in-browser AI code editor that allows users to write AI training/inference code templates in the browser, run them, and return results
- **Function Parsing:** Implemented a function-level code parser to identify inter-function dependencies and generate a directed acyclic computation graph, organized AST traversal logic, added test cases for edge cases such as nested functions and anonymous functions, and generated traceable logs for each dependency
- **Integration Testing and Monitoring:** Integrated and tested the Argo workflow deployment/runtime engine and the data scheduling aggregator in the Knative environment; monitored Pod metrics via Kubernetes and Prometheus with frontend visualization of real-time workflow metrics; achieved a 20% reduction in workflow deployment time (verified by subsequent testers)

Social Practice

Tencent Kaiwu AI Platform Professional Practice | Core Member **Jul. 2026 - Aug. 2026**

- Conducted reinforcement learning algorithm application experiments on the Tencent Kaiwu platform, mastering reward function design principles and parameter tuning mechanisms
- Authored the report *Ethical Dilemmas and Security Defenses in the Development of Artificial Intelligence* after a knowledge-base construction experiment, AI-powered literature parsing, and multi-model dialogue generation

Summer Social Practice Project | Project Leader **Jul. 2024 - Aug. 2024**

- Managed the full project lifecycle and organizational implementation, including selecting and coordinating practice locations, contacting interviewees, and planning practice content; responsible for post-research data organization
- Earned recognition as an Excellent Exhibition Project by the university

Extra-curricular Experience

- Youth League Committee Organization Department, School of Computer Science and Technology, Tongji University | Director Sep. 2025 - Sep. 2026
- Basketball Association, Tongji University Jiading Campus | Director Sep. 2025 - Sep. 2026
- Student Affairs Office, School of Computer Science and Technology, Tongji University | Student Assistant Mar. 2025 - Jul. 2025

Professional Skills

- Programming Languages: Python, C++, SQL
- Big Data & Cloud-Native Systems: Spark, Hive; Kubernetes, Knative, Argo, Prometheus
- ML/DL Frameworks & Libraries: PyTorch, NumPy, Pandas
- AI Methods & Domains: Machine Learning, Deep Learning, Transformer, Attention Mechanisms, Reinforcement Learning, NLP, Medical Image Segmentation, Causal Graph Discovery